

Lab 4 - OpenClaw Tools

Enable and test the built-in browser tool and install Firecrawl for web scraping.

Lab environment: Hostinger VPS (from Lab 1) or Docker Desktop (Windows 10/11, macOS 12+, Ubuntu 22.04+)

Prerequisite: Lab 2 completed (working AI model) + Lab 3 completed (at least one channel active). Estimated time: 30 minutes

Docker Desktop users: Prefix every openclaw command with `docker exec -it openclaw`.

Example: `docker exec -it openclaw openclaw plugins list`

Docker Desktop users: Use Ollama as the AI model for this lab. Cloud API keys (OpenAI, Groq, Gemini) have free-tier quota limits that block testing in a classroom environment. Ollama is free, runs on your machine, and has no quota.

Step 0 - Set Up Ollama Before Testing Tools (Docker Desktop)

All tools in this lab require a working AI model to process results. Set up Ollama first - it is the recommended model for Docker Desktop.

Step 0.1 - Install Ollama

Download and install from <https://ollama.com/download> (Windows). Launch the Ollama app - it runs in the system tray.

Step 0.2 - Pull a Model (PowerShell)

```
ollama pull llama3.2
```

Expected: success after download completes.

Step 0.3 - Configure OpenClaw to Use Ollama

```
docker exec -it openclaw openclaw configure --section model
```

In the wizard: select Ollama → host: `http://host.docker.internal:11434` → model: `llama3.2`

Step 0.4 - Register the Model in Config (Docker Desktop only)

The wizard sets the default but does not register the model entry. Run this to fix it:

```
docker exec openclaw sh -c 'cat > /tmp/fix-ollama.js << 'EOF''
const fs=require("fs");
const p=process.env.HOME+"/.openclaw/openclaw.json";
const c=JSON.parse(fs.readFileSync(p));
if(!c.models) c.models={};
if(!c.models.providers) c.models.providers={};
if(!c.models.providers.ollama) c.models.providers.ollama={api:"ollama",baseUrl:"http://host.docker.internal:11434",models:[]};
if(!c.models.providers.ollama.models) c.models.providers.ollama.models=[];
const exists=c.models.providers.ollama.models.find(function(m){return m.id=="llama3.2";});
if(!exists) c.models.providers.ollama.models.push({id:"llama3.2",name:"llama3.2"});
c.models.providers.ollama.baseUrl="http://host.docker.internal:11434";
fs.writeFileSync(p,JSON.stringify(c,null,2));
console.log("done");
EOF'
MSYS_NO_PATHCONV=1 docker exec openclaw node /tmp/fix-ollama.js
```

Expected output: done

Step 0.5 - Set as Default and Restart

```
docker exec -it openclaw openclaw models set ollama/llama3.2
docker restart openclaw
```

Step 0.6 - Verify the Model is Working

Send this in Telegram or WhatsApp:

```
Hello, are you working?
```

Expected: Agent replies normally. Once this works, proceed to Tool A below.

Tool A - Agent Browser (Built-in, No Setup)

The browser plugin is already installed and enabled in OpenClaw. No configuration needed.

Step A1 - Confirm Browser Plugin is Enabled

```
# VPS
openclaw plugins inspect browser

# Docker Desktop
docker exec -it openclaw openclaw plugins inspect browser
```

Expected output includes: `` Status: loaded id: browser ``

Step A2 - Test Browser via Chat

In your Telegram or WhatsApp chat with the agent, send:

```
Browse https://news.ycombinator.com and summarise the top 3 stories
```

Expected: Agent fetches the page and replies with a summary.

Tool B - Web Search (DuckDuckGo)

The DuckDuckGo search plugin is bundled but disabled by default. Enable it for free web search.

Step B1 - Enable DuckDuckGo Search

```
# VPS
openclaw plugins enable duckduckgo

# Docker Desktop
docker exec -it openclaw openclaw plugins enable duckduckgo
```

Expected output: `` Plugin duckduckgo enabled. ``

Step B2 - Test Web Search via Chat

In your chat, send:

```
Search the web: what is the weather in Singapore today?
```

Expected: Agent searches and returns current results.

Tool C - Firecrawl (Web Scraping)

Firecrawl converts websites into clean structured text for the agent to read.

Step C1 - Create a Firecrawl Account

Visit <https://www.firecrawl.dev> and sign up for a free account.

Step C2 - Get Your Firecrawl API Key

Firecrawl dashboard → API Keys → copy your key (starts with fc-).

Step C3 - Install the Firecrawl Plugin

```
# VPS
openclaw plugins install clawhub:@openclaw/firecrawl-plugin

# Docker Desktop
docker exec -it openclaw openclaw plugins install clawhub:@openclaw/firecrawl-plugin
```

Expected output: `` Installing @openclaw/firecrawl-plugin... ☐ Plugin installed. ``

Step C4 - Configure the Firecrawl API Key

```
# VPS
openclaw configure --section plugins

# Docker Desktop
docker exec -it openclaw openclaw configure --section plugins
```

Follow the prompts to enter your Firecrawl API key.

Step C5 - Test Firecrawl via Chat

In your chat, send:

```
Scrape https://docs.openclaw.ai and give me a summary
```

Expected: Agent returns a clean summary of the page.

List All Plugins

```
# VPS
openclaw plugins list

# Docker Desktop
docker exec -it openclaw openclaw plugins list
```

Verification

Check	Expected
openclaw plugins inspect browser	Status: loaded
openclaw plugins inspect duckduckgo	Status: loaded
openclaw plugins inspect firecrawl-plugin	Status: loaded
Chat: Browse https://...	Agent returns page summary
Chat: Search the web: ...	Agent returns search results
Chat: Scrape https://...	Agent returns scraped content

Troubleshooting

Symptom	Fix
Browser returns empty result	Check the URL is publicly accessible
DuckDuckGo returns no results	Try a different search query
Firecrawl install fails	Check internet connection - plugin downloads from ClawHub
Firecrawl key rejected	Key must start with fc- - re-copy from firecrawl.dev dashboard
Plugin enabled but agent not using it	Restart gateway: docker restart openclaw
Agent replies "Something went wrong"	The AI model is failing - check openclaw logs --limit 20. If all providers fail, configure Ollama (free, local) following Lab 9 Strategy 3

Reference

- Plugins: <https://docs.openclaw.ai/plugins>
- Firecrawl: <https://www.firecrawl.dev>
- ClawHub: <https://clawhub.ai>